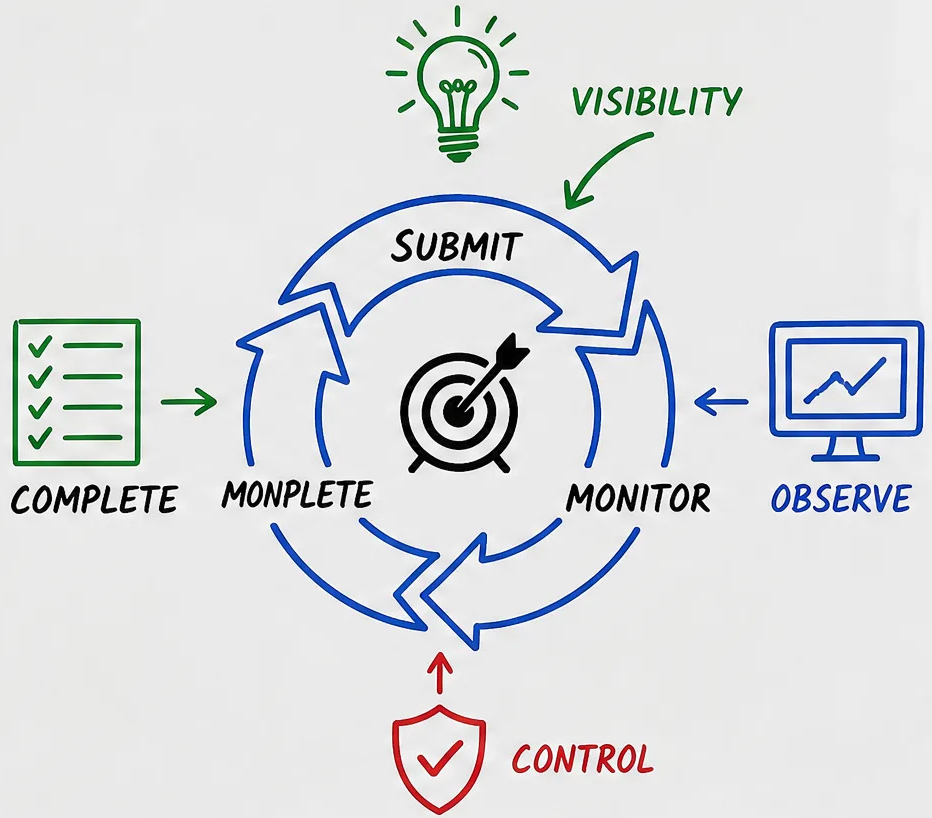
MODULE 13 — JOB MANAGEMENT

ProTech
protechtraining.com

BRING LONG WORK UNDER CONTROL.

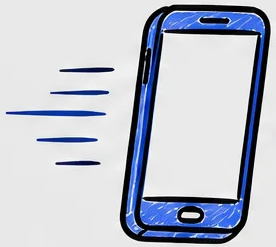


BATCH NEEDS A CONTROL PLANE, NOT HOPE.

Batch is a Business Workflow, not a Background Thread.

Master Style Reference: Realistic Whiteboard Masterclass

Time-Critical Online Request Lane

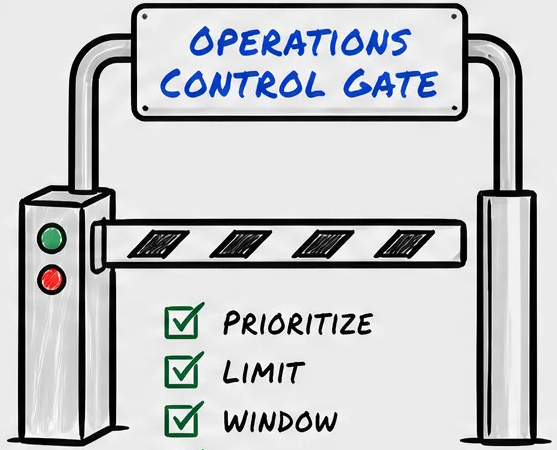


- ✓ User Request
- ✓ Low Latency
- ✓ Immediate Response



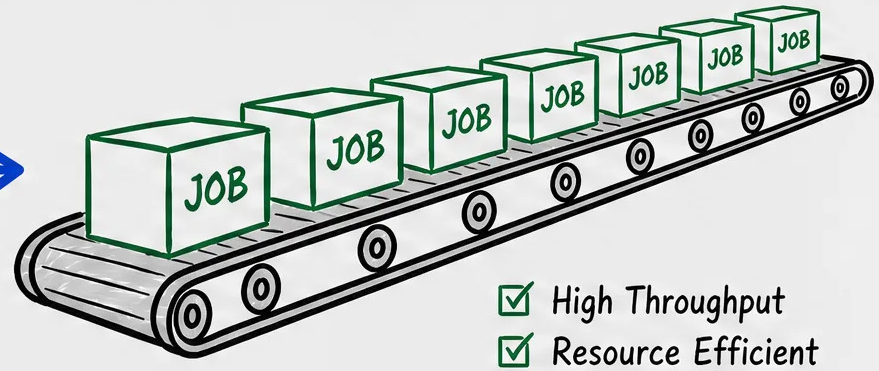
SECONDS

Make Long Work CONTROLLABLE.



- ☑ PRIORITIZE
- ☑ LIMIT
- ☑ WINDOW
- ☑ DEPENDENCIES
- ☑ MONITOR

Scheduled Long-Running Batch Conveyor



- ☑ High Throughput
- ☑ Resource Efficient
- ☑ Controlled Execution



MINUTES–HOURS (OR LONGER)

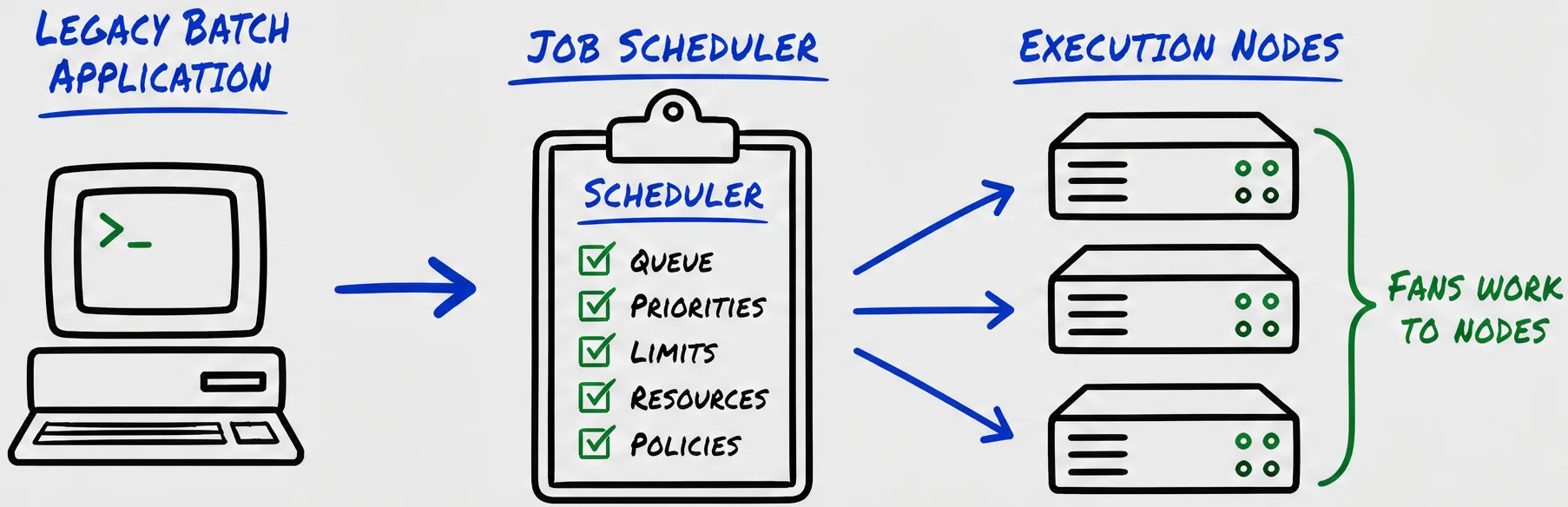


Batch needs a CONTROL PLANE, not hope.



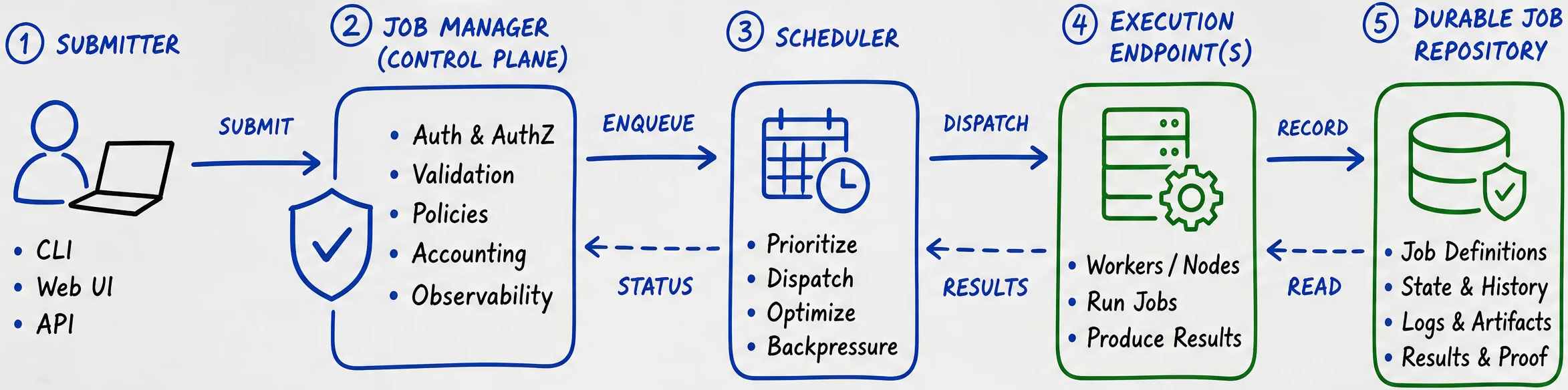
Compute Grid Is a Legacy Job Framework.

Know the Lifecycle Status.



Assess current product support before adopting.

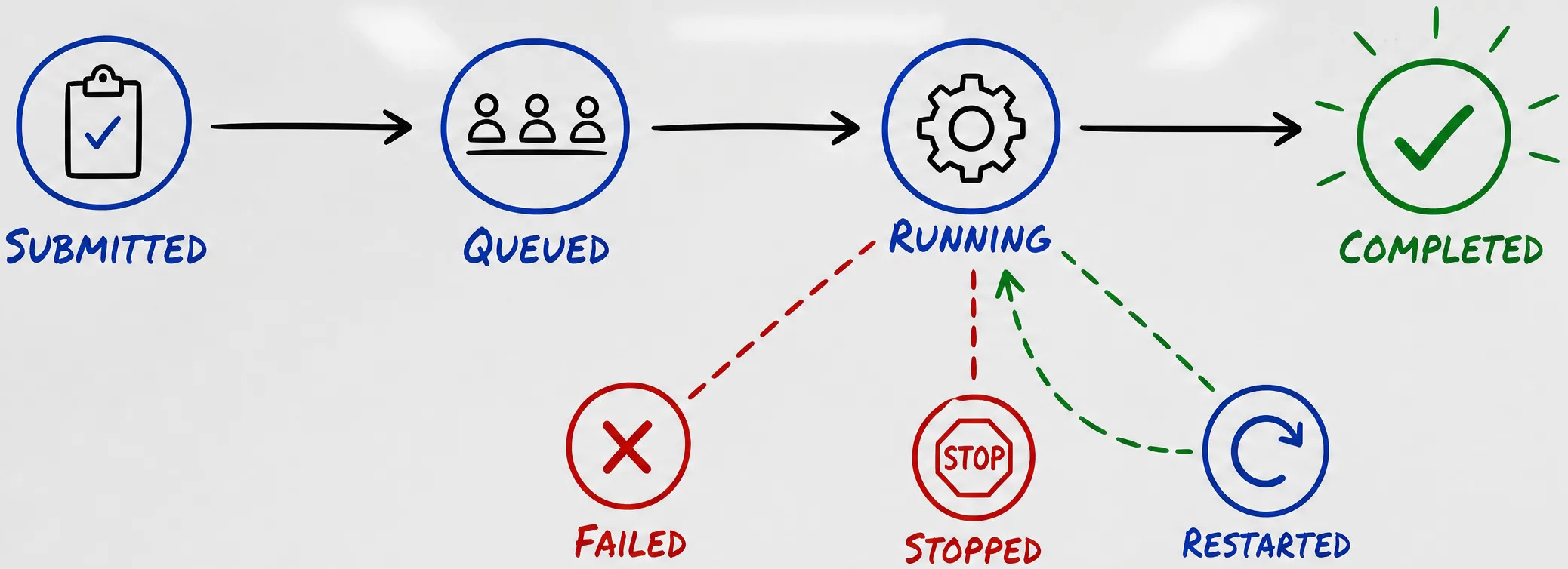
Separate the Job Control Plane From Work Execution.



Control is not execution.

A batch system needs **both** dispatch and **proof**.

Job States Make Operations Auditable.



Every Transition is Evidence.

Choose the Correct Batch Submission Pattern.

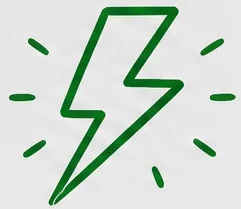
Scheduled Batch



- Time-based execution
- Predictable, repeatable schedule

Event-Triggered Workload

- Triggered by system events
- Respond in real time



Choose the Handoff.

Operator-Submitted Job



- Manually submitted by an operator
- On-demand execution

Ad Hoc Wasadmin Task



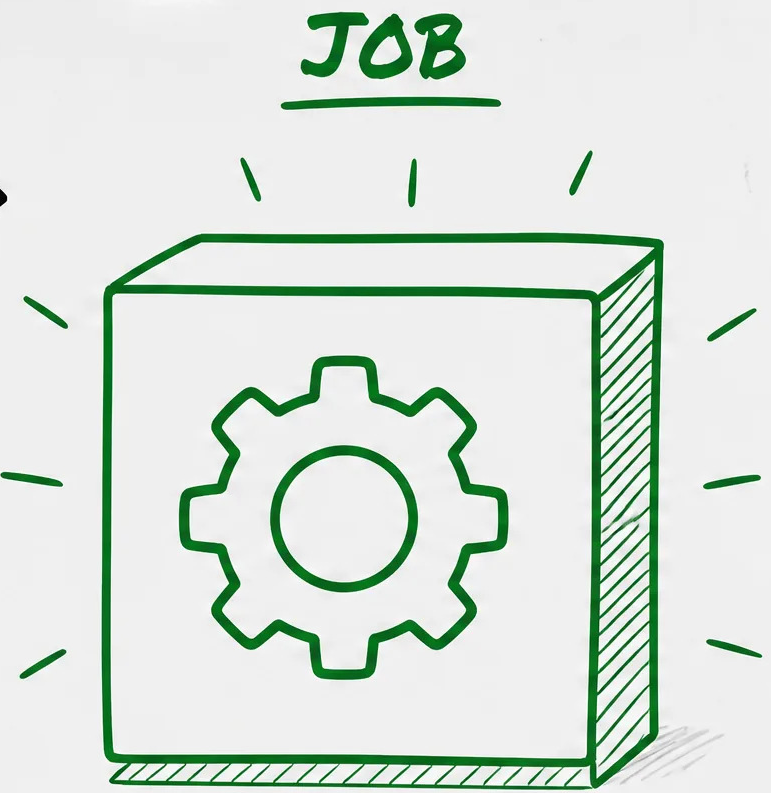
- One-off administrative task
- Immediate, targeted execution

Design Job Parameters as a Contract.

Input Manifest

- Business Date: _____
- Region: _____
- Environment: _____
- Validation Checkpoint: _____

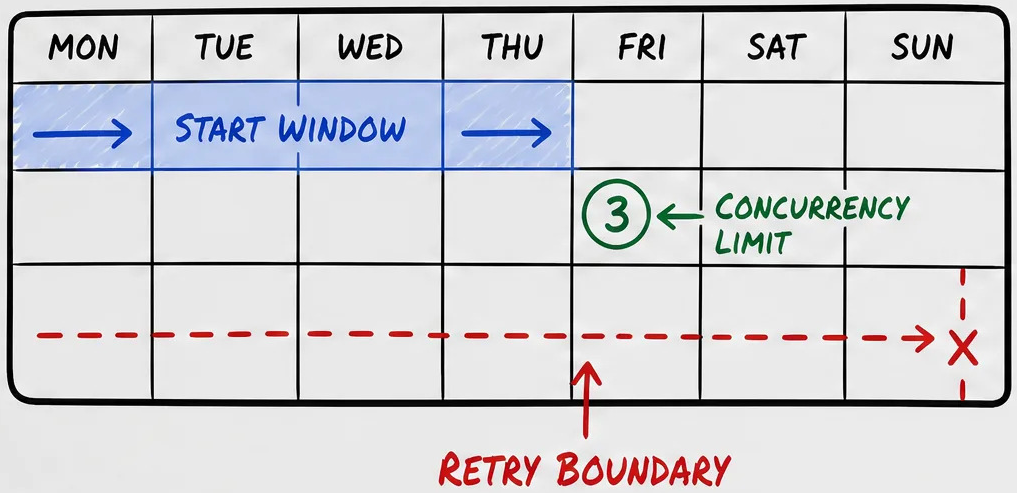
Inputs must be provable.



Schedule for Recovery, Not Just Start Time.

Time is One Control.

Job Calendar



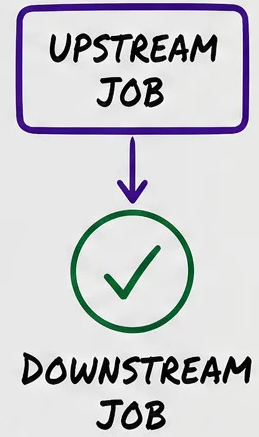
Start Window
When the job is allowed to start.

Concurrency Limit
Max number of concurrent runs.

Retry Boundary
Latest time retries are allowed.

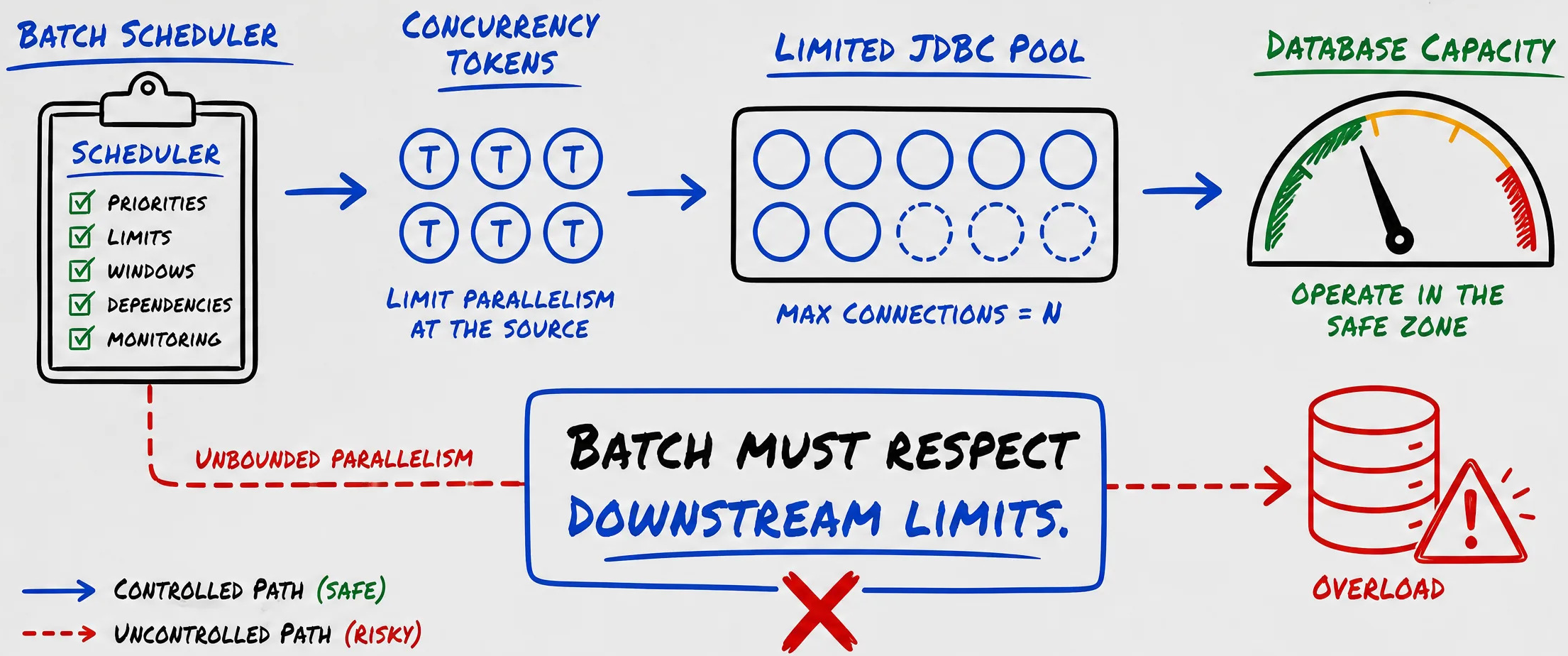
Downstream Dependency
Job waits for required downstream success.

Downstream Dependency



Design your **Schedule** to control outcomes, not just **Starts**.

Bound Concurrency Before It Reaches the Database.



Monitor the Whole Job, Not Just CPU.

Evidence Dashboard

Job State
RUNNING

Age
2h 37m

Throughput
18,450 rows/s

Error Count
3
(last 15m)

Database Waits
I/O 48%
Lock 27%
CPU 15%
Other 10%

Audit Log
10:02 Job started
10:15 Warn: Retries
10:42 Error: timeout
10:50 Retry success
10:55 Checkpoint
...

Operations needs the Story.

- See what's happening
- Understand the Impact
- Act with Confidence
- Drive Better Outcomes

Don't fly blind. **Monitor the Whole Job.**

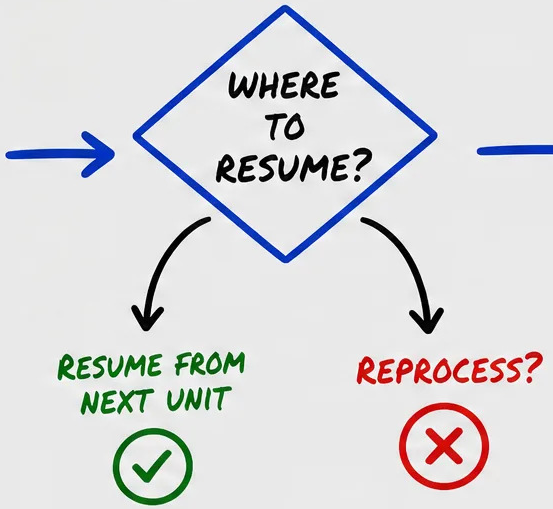
Recover From Failure Without Duplicating Work.

Checkpoint Records

WHITEBOARD CHECKPOINT

UNIT	STATUS
UNIT 101	DONE
UNIT 102	DONE
UNIT 103	DONE
UNIT 104	NEXT
⋮	

Restart Decision



Idempotence Shield



Rollback If Needed



ROLL BACK TO LAST SAFE CHECKPOINT

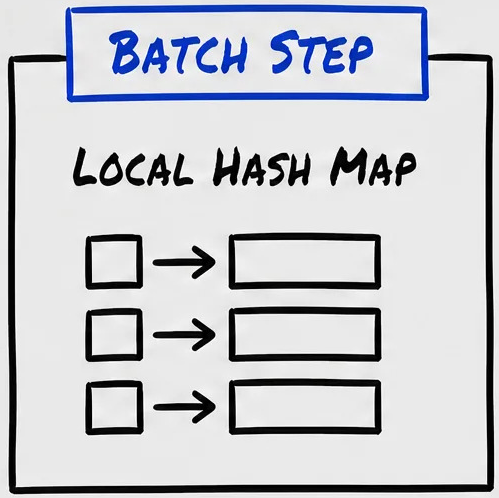


Resume Safely.



Checkpoint Data Instead of Memory.

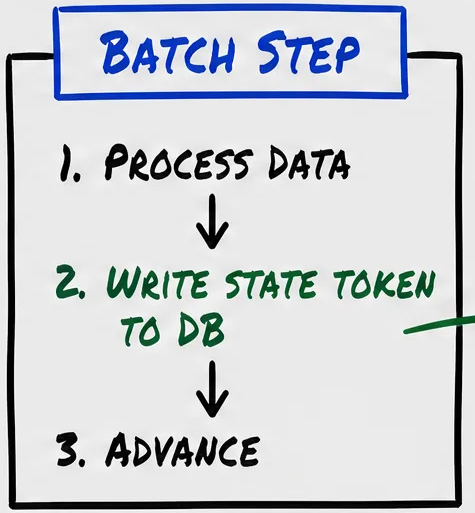
Step 1: In-Memory State (Local Hash Map)





State **Must Survive** the JVM.

Step 2: Checkpoint State Before Advancing



State Table

step_id	state_token
42	8f3a7c...
...	



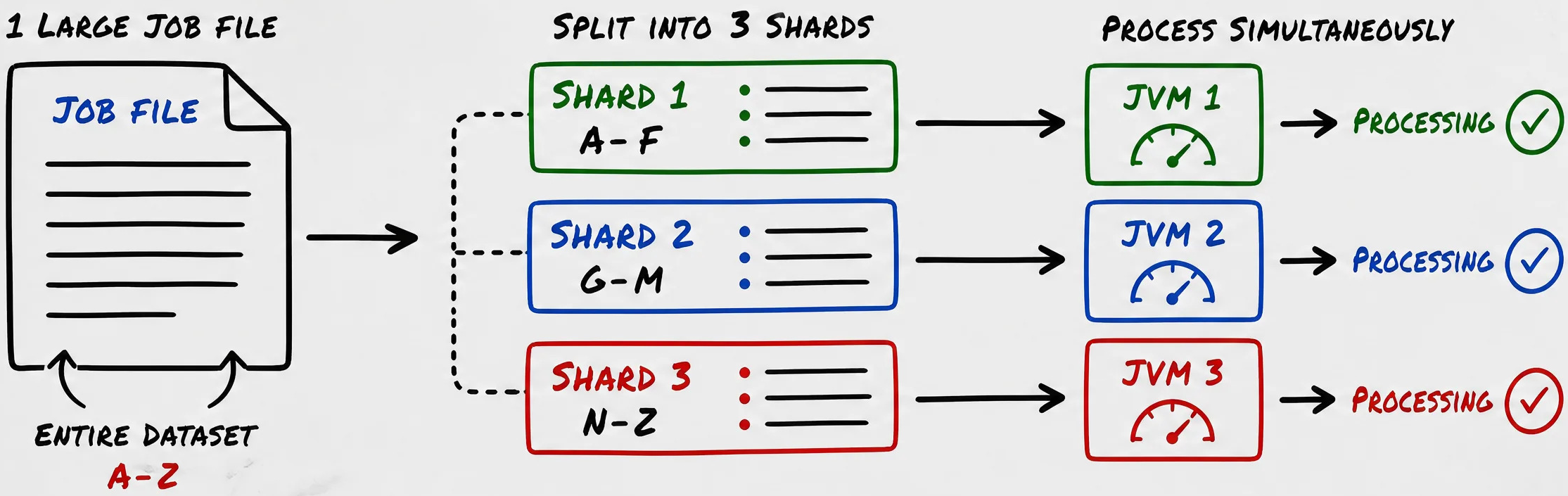
In-memory state is **lost** on failure.



State is **persisted**. Recoverable. Reliable.

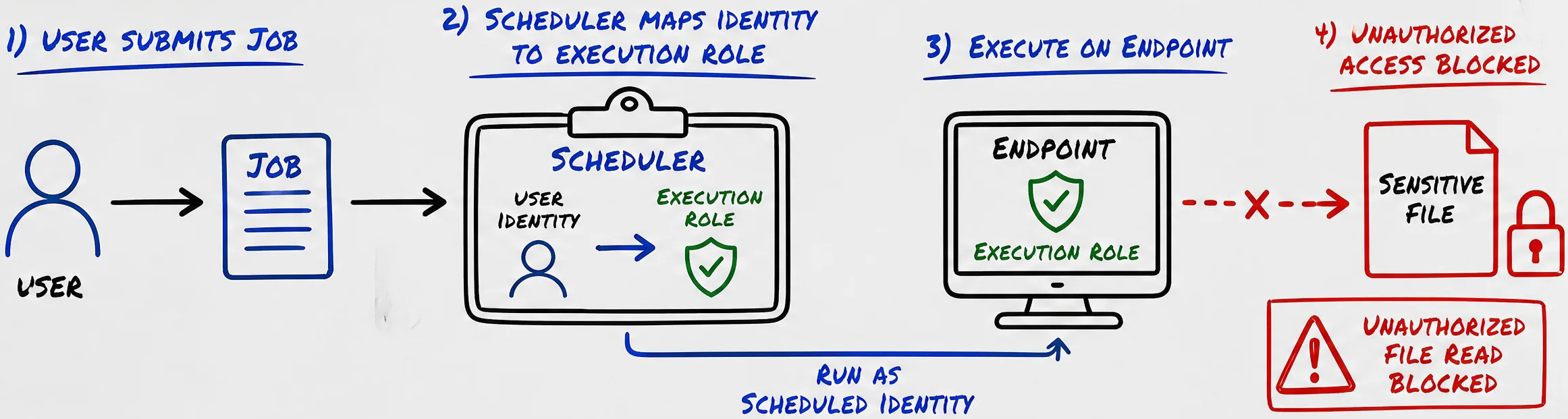
Scale Batch by Partitioning, Not Just Cloning.

Split the DATA, not the CODE.



Secure Batch Execution Prevents Escalation.

Run as the Scheduled Identity.



Right identity. Limited access. No escalation.

Lab: Dispatch and Control a Long Job

Submission Command

```
$ sbatch long_job.sh
```

Job Log

```
[10:15:01] JOB 12345 STARTED
[10:17:42] CHECKPOINT WRITTEN
[10:18:05] JOB PAUSED
[10:45:12] JOB RESUMED
[11:12:37] CHECKPOINT WRITTEN
...
```

Database Checkpoint Verification

- ☑ BEFORE PAUSE CHECKPOINT FOUND
- ☑ NO NEW CHECKPOINTS DURING PAUSE
- ☑ AFTER RESUME CHECKPOINT FOUND
- ☑ SEQUENCE VALID (CONTINUITY PROVEN)



Prove the Pause.

PAUSE

Recap: Bring Batch Under Management.

1 Submit

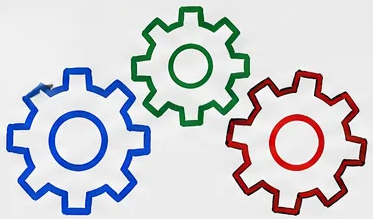


2 Dispatch



Prioritize & dispatch

3 Execute



Run • Monitor • Control

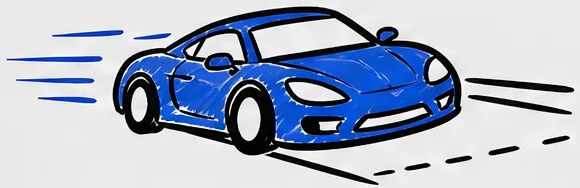
4 Checkpoint



- Status
- Results
- Audit trail
- Notifications

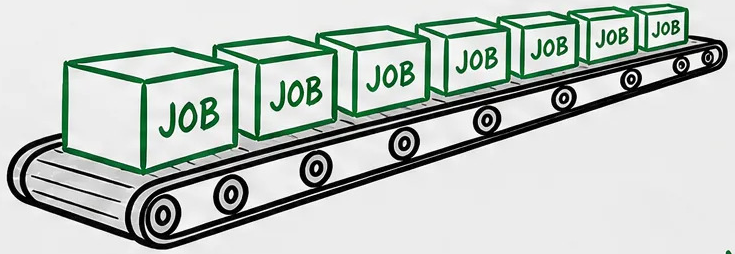
One operations standard.

Online lane



- Fast
- Responsive
- Real-time

Batch lane



- Reliable
- Scheduled
- Scalable



One standard. Two lanes. Complete control.

